

4.2.17

Find k such that $\text{Nul } A$ is a subspace of $\mathbb{R}^k$, and such that $\text{Col } A$ is a subspace of $\mathbb{R}^k$.

$$A = \begin{bmatrix} 2 & -8 \\ -1 & 4 \\ 1 & -4 \end{bmatrix} \quad \begin{array}{l} \text{Nul } A \text{ is a subspace of } \mathbb{R}^n - n=k=2 \\ \text{Col } A \text{ is a subspace of } \mathbb{R}^m - m=k=3 \end{array}$$

$m \times n$
 3×2

4.2.19

$$a: \text{Nul}(A) \subseteq \mathbb{R}^n, k=5$$

$$b: \text{Col}(A) \subseteq \mathbb{R}^m, k=2$$

$$A = \begin{bmatrix} 4 & 5 & -2 & 6 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$m \times n$
 2×5

4.2.23

Let $A = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$, and $\vec{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine if $\vec{w}$ is in $\text{Col } A$. Is $\vec{w}$ in $\text{Nul } A$?

Solution

To determine if $\vec{w}$ is in $\text{Nul } A$, we simply compute

$A\vec{w}$, if it equals $\vec{0}$, $\vec{w}$ is in $\text{Nul } A$.

$$A\vec{w} = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -12 + 12 \\ -6 + 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\vec{w}$ is in $\text{Nul } A$.

To determine if $\vec{w}$ is in $\text{Col } A$, we perform row operations on $[A \mid \vec{w}]$

$$[A \mid \vec{w}] = \left[\begin{array}{cc|c} -6 & 12 & 2 \\ -3 & 6 & 1 \end{array} \right]$$

$$r_1 \leftarrow r_1 \cdot (-1/6), r_2 \leftarrow r_2 \cdot (-1/3), r_2 \leftrightarrow r_1$$

$$\left[\begin{array}{cc|c} 1 & -2 & -1/3 \\ 1 & -2 & -1/3 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & -1/3 \\ 0 & 0 & 0 \end{array} \right]$$

$$x_1 - 2x_2 = -1/3, x_2 = s$$

$$x_1 = -1/3 + 2s$$

$$\vec{x} = \begin{bmatrix} 2s - 1/3 \\ s \end{bmatrix} = \begin{bmatrix} -1/3 \\ 0 \end{bmatrix} + s \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\vec{u}}$$

$$2 \begin{bmatrix} -6 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -10 \\ -5 \end{bmatrix}$$

$$2 \begin{bmatrix} -6 \\ -3 \end{bmatrix} + 7 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$\vec{w}$ is in $\text{Col } A$

4.2.24

Let $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if $\vec{w}$ is in Col A or Nul A .

Solution

$$\text{Perform } A\vec{w} \quad \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -16 - 2 + 18 \\ 12 + 4 - 18 \\ 8 + 0 - 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \boxed{\vec{w} \text{ is in Nul } A}$$

Perform row operations on $[A | \vec{w}]$

$$\left[\begin{array}{ccc|c} -8 & -2 & -9 & 2 \\ 6 & 4 & 8 & 1 \\ 4 & 0 & 4 & -2 \end{array} \right]$$

$$\begin{array}{l} r_3 \leftarrow r_3 + r_2 \\ r_2 \leftarrow r_2 \cdot \frac{1}{2} \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1/2 \\ 0 & 1 & 1/2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$r_1 \leftrightarrow r_3$$

$$r_1 \leftarrow r_1 \cdot \frac{1}{4}$$

$$r_2 \leftarrow r_2 \cdot \frac{1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1/2 \\ 3 & 2 & 4 & 1/2 \\ -8 & -2 & -9 & 2 \end{array} \right]$$

$$-\frac{1}{2} \begin{bmatrix} -8 \\ 6 \\ 4 \end{bmatrix} + 1 \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\boxed{\vec{w} \text{ is in Col } A}$$

$$r_2 \leftarrow r_2 - 3r_1$$

$$r_3 \leftarrow r_3 + 8r_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1/2 \\ 0 & 2 & 1 & 2 \\ 0 & -2 & -1 & -2 \end{array} \right]$$